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Non-conventional Water Resources for Irrigation: a Critical Scientometric Review of Global Research Trends, Evidence Gaps, and Priorities (2020–2024)

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Abstract

Water scarcity has accelerated scientific interest in non-conventional water resources for irrigation. This study combined performance analysis, science mapping, and a critical interpretive synthesis of Web of Science literature published from 2020 to 2024 to evaluate not only the structure of the field, but also the strength, limitations, and practical implications of the available evidence. Publication volume increased from 122 articles in 2020 to 174 in 2024, with a peak of 190 in 2022, and collaboration networks were concentrated in Brazil, China, the Middle East, and Europe. However, bibliometric visibility did not equate to evidence maturity. The literature consistently reports potential benefits, including freshwater substitution, nutrient recovery, and greater resilience to drought, alongside unresolved risks related to salinity and sodicity, contaminants of emerging concern, pathogens, antibiotic-resistance determinants, and groundwater contamination. The critical synthesis identified five major limitations: predominance of short-term and site-specific experiments; inconsistent characterization of water, soil, and crops; scarce long-term, multi-compartment monitoring; fragmented regulatory thresholds; and limited techno-economic and adoption evidence. We therefore propose priorities centered on harmonized reporting, multi-season field trials, mixture-based and One Health risk assessment, fit-for-purpose treatment, validation of digital monitoring under operational conditions, lifecycle assessment, and context-specific regulation. Non-conventional water should not be framed as universally sustainable; its performance is conditional on source quality, treatment, crop sensitivity, soil buffering capacity, irrigation design, monitoring, and governance. The field is scientifically active, but its operational readiness remains uneven.

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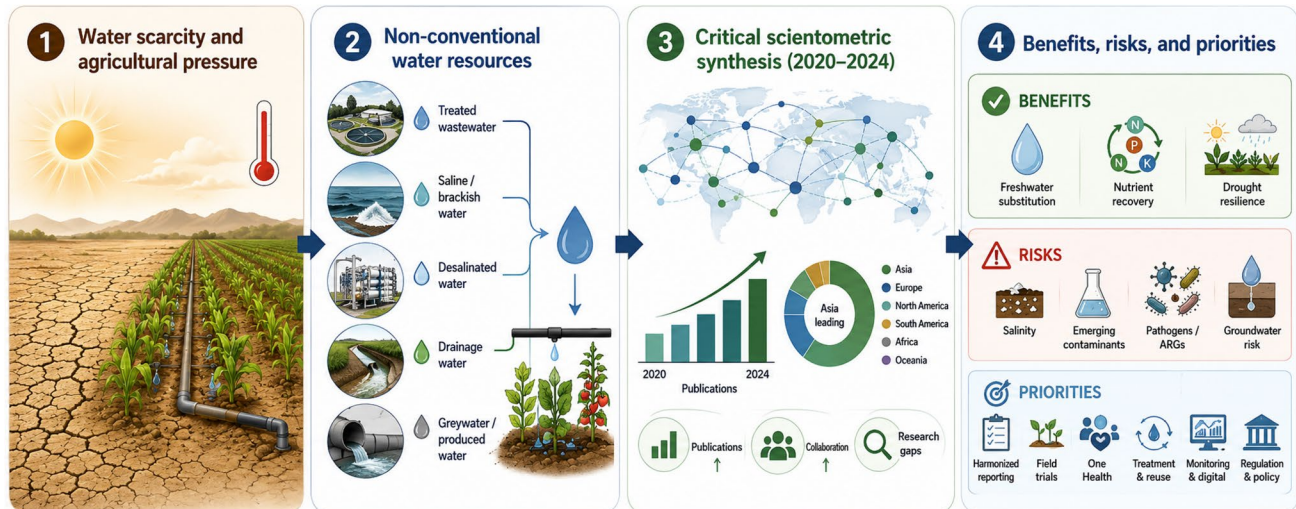
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Graphical Abstract



Highlights

- Global research on non-conventional irrigation water expanded from 2020 to 2024.
- Evidence remains fragmented across water types, crops, soils, and climates.
- Benefits coexist with salinity, contaminants, pathogens, and soil degradation risks.
- One Health assessments and multi-season field trials are urgently needed.
- Harmonized monitoring, treatment, and regulation can support safe water reuse.

Keywords Water reuse · Saline water · Wastewater irrigation · Emerging contaminants · Research gaps · Scientometrics

Introduction

Freshwater scarcity increasingly constrains agricultural production and intensifies competition among food production, cities, industry, and ecosystems. Agriculture accounts for approximately 72% of global freshwater withdrawals, so reduced water availability or reliability directly affects food-system resilience [39]. Climate variability, recurrent drought, groundwater depletion, salinization, and rising demand have increased interest in sources that can supplement conventional freshwater supplies, particularly where irrigation already operates close to hydrological limits.

The pressure is especially acute in arid and semi-arid regions, but it is not restricted to them. In Europe, reclaimed water is being incorporated into climate-adaptation strategies for irrigated agriculture [15], while countries in the Middle East and North Africa face chronic water deficits together with concerns about food safety, farmer acceptance, and monitoring capacity [11]. Similar pressures have stimulated research on saline and brackish water, desalinated water, drainage water, greywater, and treated municipal or industrial wastewater across Asia and Latin America. These sources are not agronomically or environmentally equivalent: wastewater reuse may recover water and nutrients

while introducing salts, trace elements, pathogens, antibiotic-resistance determinants, pharmaceuticals, and other contaminants, whereas saline or brackish water is dominated by osmotic stress, ion toxicity, sodicity, and long-term soil-structure effects. Evidence on agriculture-derived groundwater contamination further illustrates why alternative-water use should be evaluated as an exposure pathway rather than only as a water-supply intervention [36].

The research gap is therefore twofold. First, the literature is expanding rapidly but remains heterogeneous in water-source definitions, experimental duration, water-soil-crop characterization, analytical endpoints, and regulatory context. Second, conventional scientometric mapping can describe publication growth and collaboration without establishing whether highly visible topics are supported by transferable, long-term evidence. The novelty of this review is the integration of performance analysis and science mapping with a critical interpretive synthesis focused on evidence maturity and implementation. Specifically, we aimed to (i) map global research on non-conventional water use for irrigation during 2020–2024; (ii) critically assess evidence on agronomic performance, environmental and health risks, treatment, monitoring, and governance; and (iii) identify priority gaps and conditions required for safe, context-specific implementation.

Materials and Methods

Data Source, Eligibility Criteria, and Search Strategy

Bibliographic records were retrieved from the Web of Science (WoS) Core Collection in plain text (.txt) format, covering titles, abstracts, author keywords, and Keywords Plus, aiming to maximize the retrieval of relevant articles. Eligibility was defined by publication year (2020–2024), document type (Article or Review Article), and matching the Topic-field query described below; no geographic filter was applied and the retrieved records were standardized and manually screened, and duplicate entries were prior to analysis. The search was executed on 20 November 2025. The resulting analytical corpus comprised 763 records: 122 (2020), 129 (2021), 190 (2022), 148 (2023), and 174 (2024).

The Topic search, combined irrigation, non-conventional water-source, and agricultural-context terms: TS=((irrigat* OR "agricultural irrigat" OR "crop irrigat" OR "farm irrigat") AND ("reclaimed water" OR "wastewater reuse" OR greywater OR "saline water" OR "non-conventional water") AND (crop* OR soil* OR "food production" OR "plant growth")). The wildcard (*) captured lexical variants, OR combined synonymous terms, and AND linked the three conceptual blocks.

Bibliometric analyses were conducted using Bibliometrix v3.0 in the RStudio environment (R version 4.5.2; RStudio version 2025.09.2) and VOSviewer v1.6.18 [42].

Bibliometrix was used for annual production, Web of Science research areas, productivity and citation rankings, factor analysis, document coupling, and thematic mapping. VOSviewer was used for journal co-citation, author/institution/country co-authorship, and keyword co-occurrence networks.

For network maps, the minimum thresholds were ≥ 5 documents for countries and institutions, ≥ 3 citations for journals and articles, and ≥ 5 occurrences for keywords. Total link strength quantified network connectivity. Figure 1 was regenerated from annual record counts and mean citations per article using Python 3.13.5 and Matplotlib 3.10.8. The plotted values were 122/6 (2020), 129/5 (2021), 190/4 (2022), 148/3 (2023), and 174/2 (2024), where each pair represents documents/mean citations per article. An ordinary least-squares fit to annual publication counts is shown only as a descriptive trend, not as an inferential test.

Critical Interpretive Synthesis and Gap Identification

Bibliometric outputs were complemented by a critical interpretive synthesis organized around six decision domains: source type and quality; agronomic performance; soil, crop, groundwater, and food-safety risks; treatment and irrigation technologies; monitoring and regulation; and socioeconomic feasibility and adoption. Evidence was compared across water-soil-crop contexts, study duration and scale, and degree

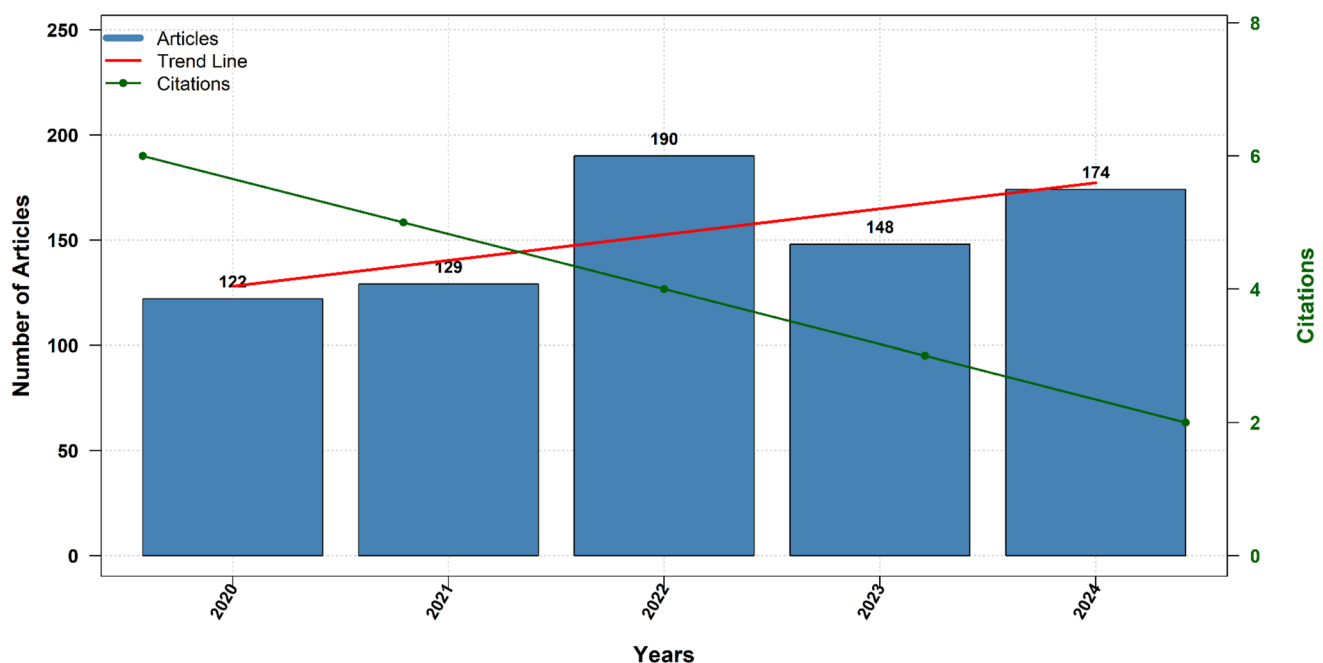


Fig. 1 Annual scientific production and mean citations per article for research on non-conventional water resources for irrigation (2020–2024). Bars show annual document counts (122, 129, 190, 148, and 174), and points show mean citations per article (6, 5, 4, 3, and 2). The

straight line is an ordinary least-squares trend fitted to annual publication counts and is shown only as a descriptive aid. Figure generated in Python 3.13.5 using Matplotlib 3.10.8

of operational validation, with emphasis on convergent findings and contradictions rather than a catalogue of studies.

A knowledge gap was considered relevant when a recurrent risk or management need was visible in the thematic structure but was supported mainly by short-duration, single-site, laboratory, greenhouse, modelling, or review evidence. The study did not conduct a formal risk-of-bias assessment or meta-analysis; consequently, the recommendations are interpreted as an evidence-informed research agenda rather than pooled estimates of efficacy or safety.

Results and Discussion

Temporal Trend and Volume of Global Scientific Production on Non-conventional Resources in Irrigation Over the last 5 Years

The corpus shows an overall increase in scientific attention to non-conventional irrigation water during 2020–2024 (Fig. 1), but the pattern is not monotonic. The five-year series is sensitive to annual publication cycles, and the fitted line is used descriptively rather than as evidence of sustained statistical growth.

Annual output rose from 122 records in 2020 to 174 in 2024 (+42.6%), with 129 records in 2021, a peak of 190 in 2022, and 148 in 2023. The uneven trajectory supports continued scientific relevance but not uninterrupted consolidation; the more important question is whether publication growth has produced transferable, long-term evidence.

Average citations per article declined across the period, primarily because recent publications had less time to accumulate citations [49]. Consequently, citation counts and averages were treated as indicators of visibility, not as direct measures of scientific quality. Comparisons among years, journals, authors, institutions, or countries must also consider publication age, document type, field size, collaboration patterns, language, and database coverage.

Study of Journals, Authors, Institutions, and Countries

Journal-level indicators showed complementary patterns of productivity and citation visibility (Fig. 2). Water led the corpus with 73 documents, whereas *Science of The Total Environment* and *Sustainability* displayed higher mean citation values; *Agricultural Water Management* and *Agronomy-Basel* combined comparatively high productivity and visibility. Citation averages were interpreted as dissemination indicators rather than measures of methodological quality because they depend on publication age, document type, field size, and citation window.

The co-citation network identifies journals that are frequently cited together and therefore occupy overlapping intellectual domains. *Science of The Total Environment*, *Sustainability*, and *Agricultural Water Management* form cross-cutting nodes linking water quality, treatment, soil processes, and crop response, whereas more peripheral regional journals reflect specialization rather than lower scientific relevance. The temporal overlay indicates relatively recent citation activity but should not be interpreted as a forecast of future influence.

At the author level, de Lima, G. S. and Gheyi, H. R. led publication volume and total citations, whereas authors with fewer papers sometimes showed higher mean citations (Fig. 3). Cui, B., for example, combined eight publications with a comparatively high mean, illustrating how a small number of highly cited papers can inflate average-based indicators.

The co-authorship and temporal maps are most useful for identifying collaboration communities, recurrent contributors, and recent participation. Node size, network position, and citation density were not treated as individual quality scores because they are sensitive to publication year, journal visibility, topic, collaboration pattern, and denominator size.

Institutional co-authorship revealed distinct regional hubs (Fig. 4). Brazilian institutions formed a dense cluster led by the Federal University of Campina Grande (UFCG) and linked to UFRSA, UFC, UFPE, and UFV; Chinese institutions clustered around the Chinese Academy of Sciences and associated universities; and Middle Eastern/North African networks included King Saud University, Al-Azhar University, Mansoura University, Alexandria University, and other regional institutions. Euro-American organizations, including universities in the United States, Portugal, Italy, and CSIC in Spain, acted as interregional bridges.

The temporal overlay suggests relatively recent activity among several Asian and Middle Eastern institutions alongside sustained Brazilian participation, but the five-year window is too short to infer a durable shift in scientific leadership. In the performance panel, UFCG had the highest publication count and total citations, whereas high mean citation values for smaller organizations should be interpreted cautiously because small denominators amplify the effect of a few highly cited papers.

Country-level patterns similarly separated productivity from citation visibility (Fig. 5). China and Brazil led publication output, while Italy, Pakistan, Spain, Egypt, and Saudi Arabia showed comparatively higher mean citation values. The collaboration network highlighted China, the United States, Italy, Spain, India, and Brazil as prominent nodes and showed substantial connectivity among water-stressed countries in the Middle East and South Asia.

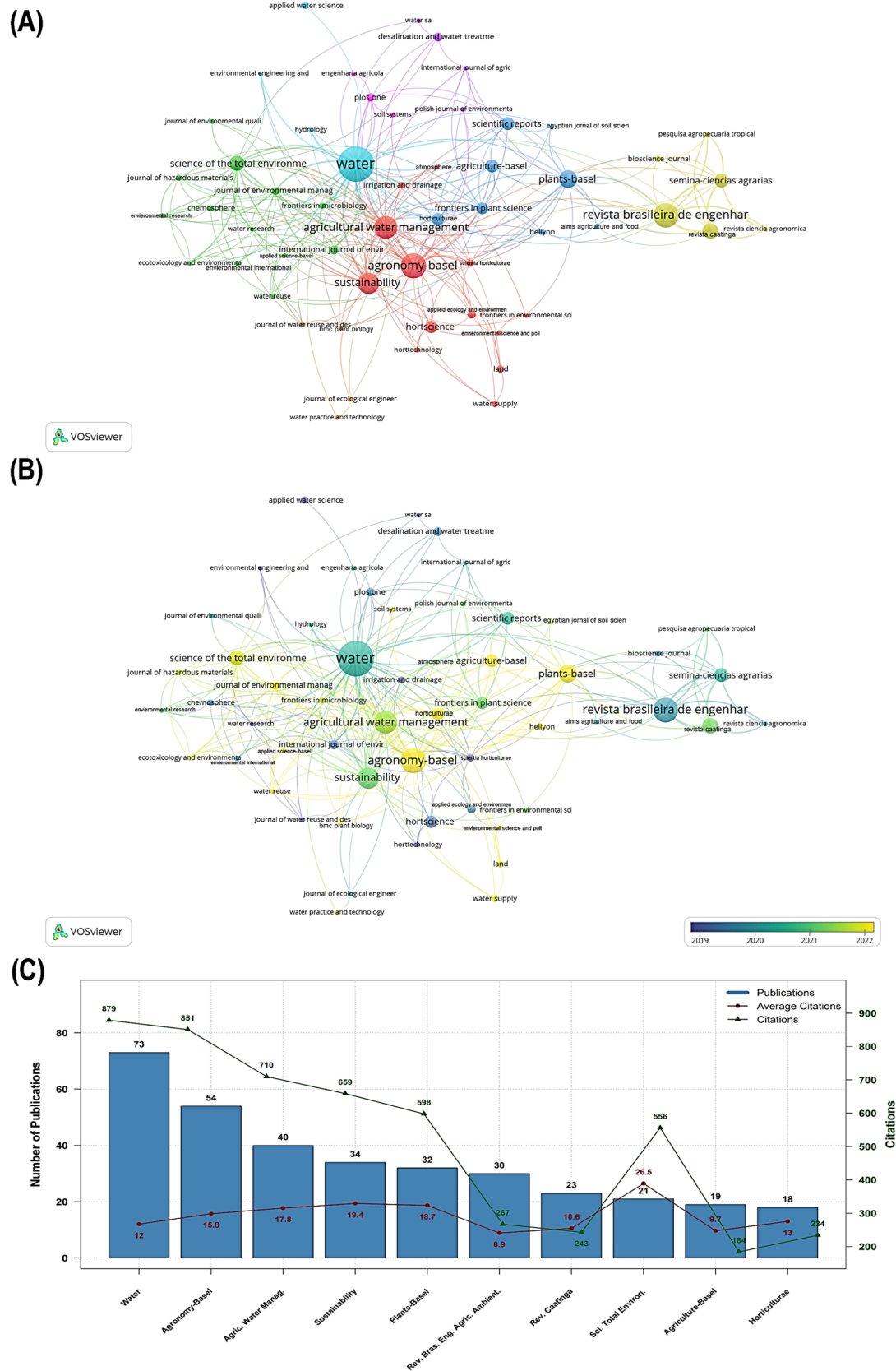


Fig. 2 Journal-level co-citation structure and publication performance in research on non-conventional water resources for irrigation. **A** Journal co-citation network and thematic clusters. **B** Temporal overlay

based on average citation year. **C** Leading journals by number of documents, total citations, and mean citations per article

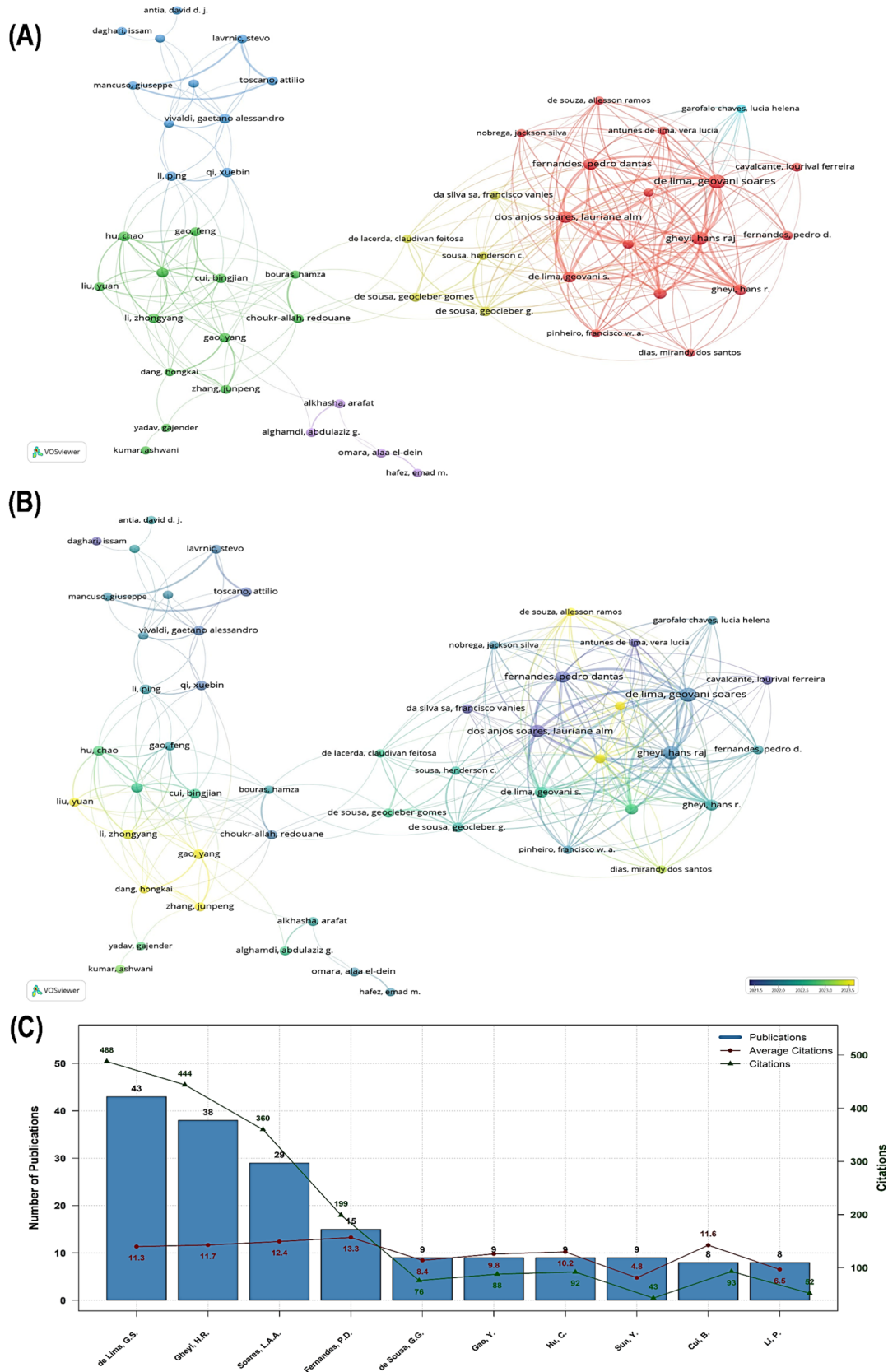


Fig. 3 Author collaboration structure and publication performance in research on non-conventional water resources for irrigation. **A** Co-authorship network and collaboration clusters. **B** Temporal overlay

based on average publication year. **C** Leading authors by number of documents, total citations, and mean citations per article

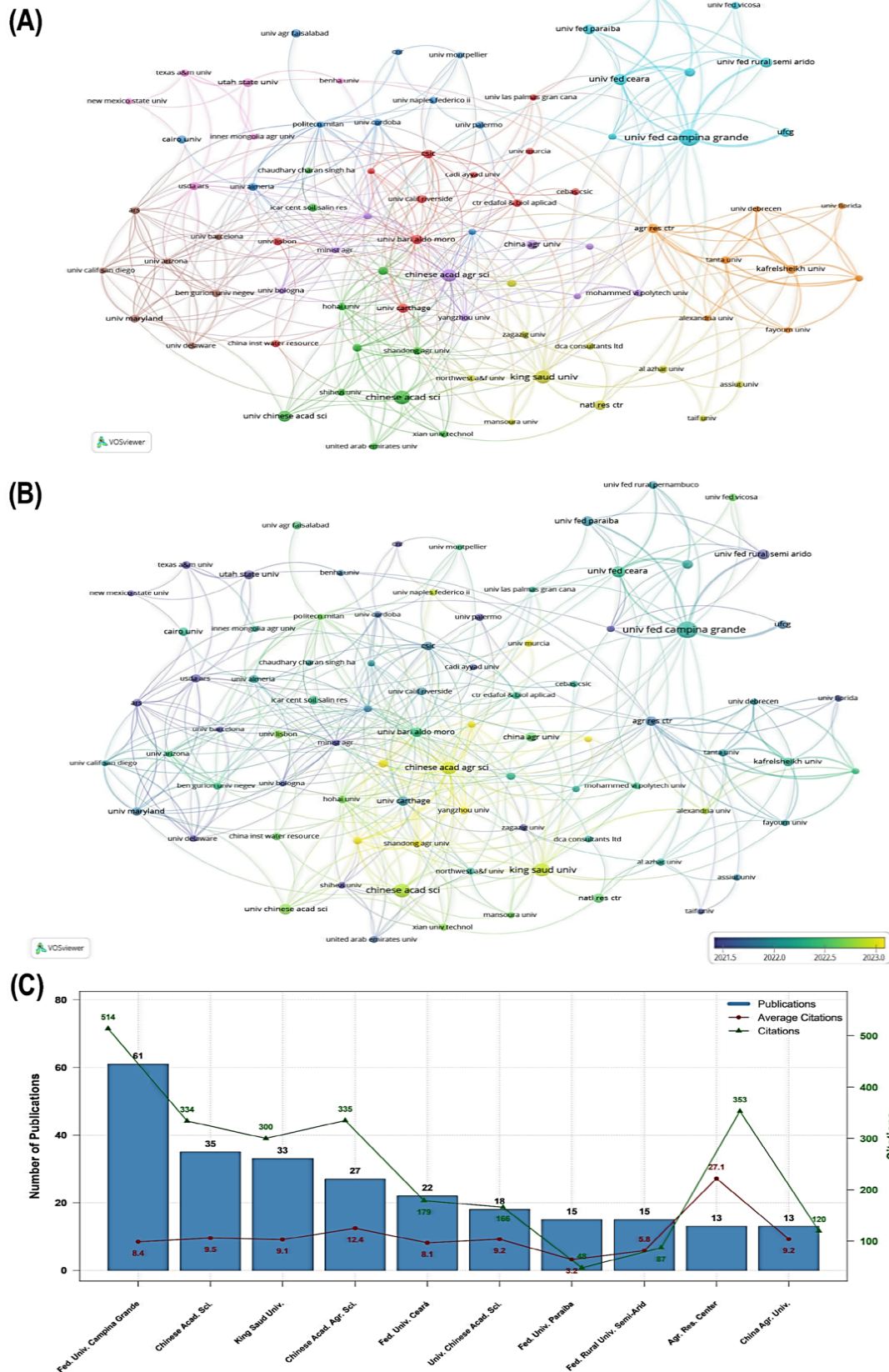


Fig. 4 Institutional collaboration and publication performance in research on non-conventional water resources for irrigation. **A** Co-authorship network among organizations. **B** Temporal overlay based

on average publication year. **C** Leading organizations by documents, total citations, and mean citations per article

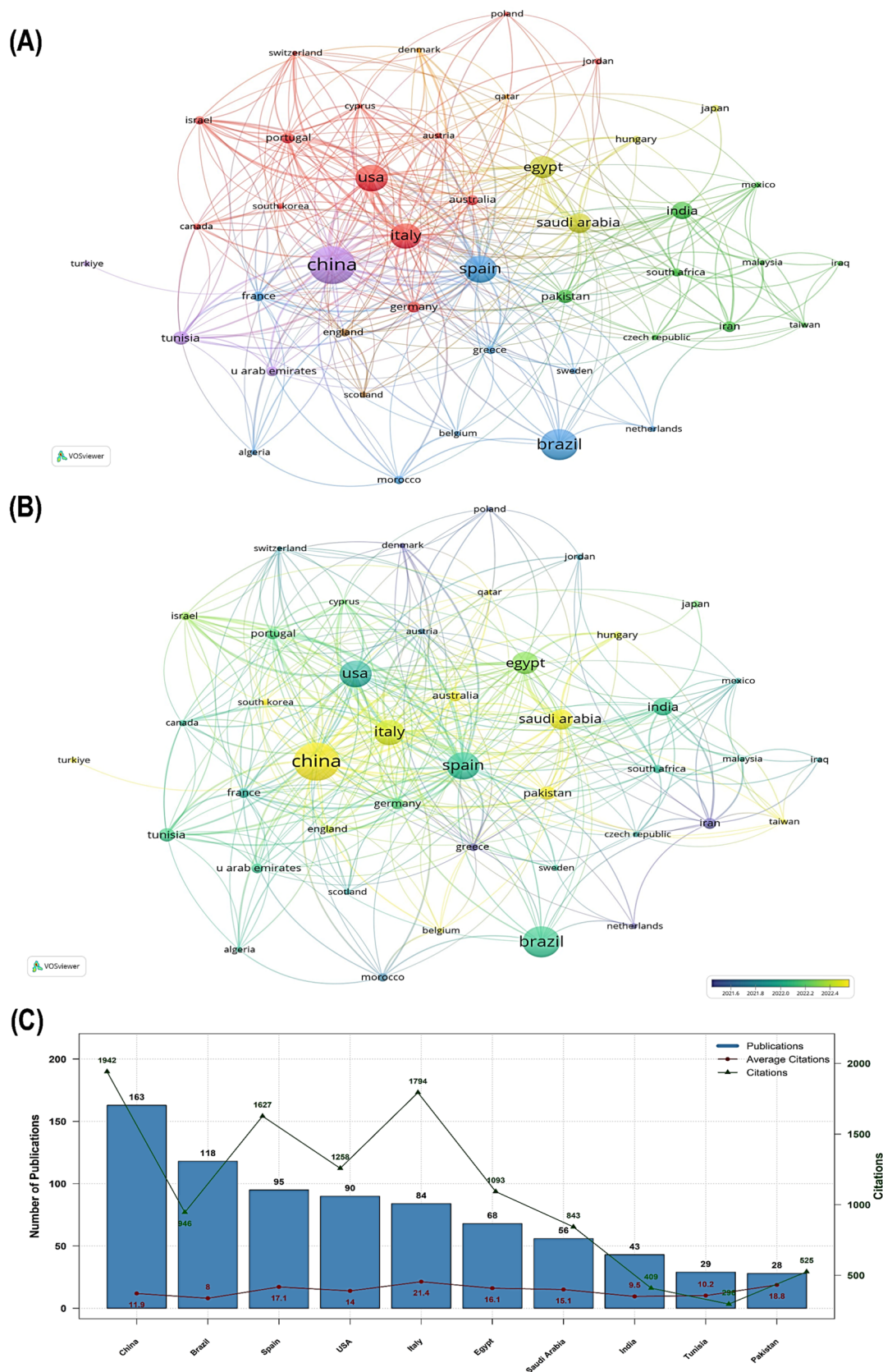


Fig. 5 Country-level collaboration and publication performance in research on non-conventional water resources for irrigation. **A** International co-authorship network. **B** Temporal overlay based on average

publication year. **C** Leading countries by documents, total citations, and mean citations per article

These differences describe publication and collaboration profiles, not national research efficiency. Citation averages and network centrality are influenced by publication age, field composition, international co-authorship, journal distribution, and database coverage. The temporal overlay therefore supports cautious inference of geographic diversification, not a causal or quality ranking.

Most Cited Articles in the Research Area

The highly cited literature is dominated by broad reviews, models, and early-window papers, which generally address wider audiences and have had more time to accumulate citations (Fig. 6). Ingrao et al. [16], the most cited paper in the dataset (275 citations), framed responses to agricultural water scarcity across policy, socioeconomic, and engineering dimensions. Citation rank is therefore treated as agenda-setting visibility rather than a direct measure of empirical evidence strength.

Among the remaining highly cited works, Ungureanu et al. [41] reviewed wastewater reuse under water scarcity and emphasized nutrient benefits together with long-term salinity, metal, and pathogen risks. Rizzo et al. [32] focused on advanced treatment trains for municipal wastewater reuse and concluded that no single technology universally controls contaminants of emerging concern and antibiotic-resistance determinants. Nikolaou et al. [28] emphasized irrigation efficiency, precision scheduling, subsurface drip irrigation, and regulated deficit irrigation as complementary water-saving strategies.

Helmecke et al. [14] highlighted regulatory gaps for organic micropollutants and plant uptake, Hashem and Qi [13] summarized trade-offs among nutrient supply, soil fertility, heavy metals, microbiological hazards, and long-term reuse, and Yang et al. [47] showed how biochar and irrigation regime can mitigate combined drought-salinity stress. Together, these studies illustrate a shift from simple water substitution toward integrated management of water quality, soil processes, crop response, and treatment.

More technology- and policy-oriented papers also ranked highly. Zhao et al. [51] demonstrated a solar desalination-irrigation system using a graphene-based sponge, whereas Hristov et al. [15] modelled the potential for treated-water reuse to reduce European agricultural water scarcity. Compagni et al. [8] integrated water-quality and crop-uptake modelling and identified health-risk concerns for selected contaminants of emerging concern, while showing that irrigation configuration can modify exposure.

Taken together, the most-cited works converge on three messages: no single treatment train is universally sufficient; agronomic benefits depend on source-specific management; and risk assessment must extend beyond salinity and conventional microbiological indicators to emerging contaminants, antibiotic resistance, and cumulative exposure. However, the predominance of reviews, models, and technology demonstrations reveals a translation gap. Future work should prioritize multi-season field validation that simultaneously measures yield, cumulative loads, treatment reliability, energy demand, costs, residuals, and farm-operational feasibility.

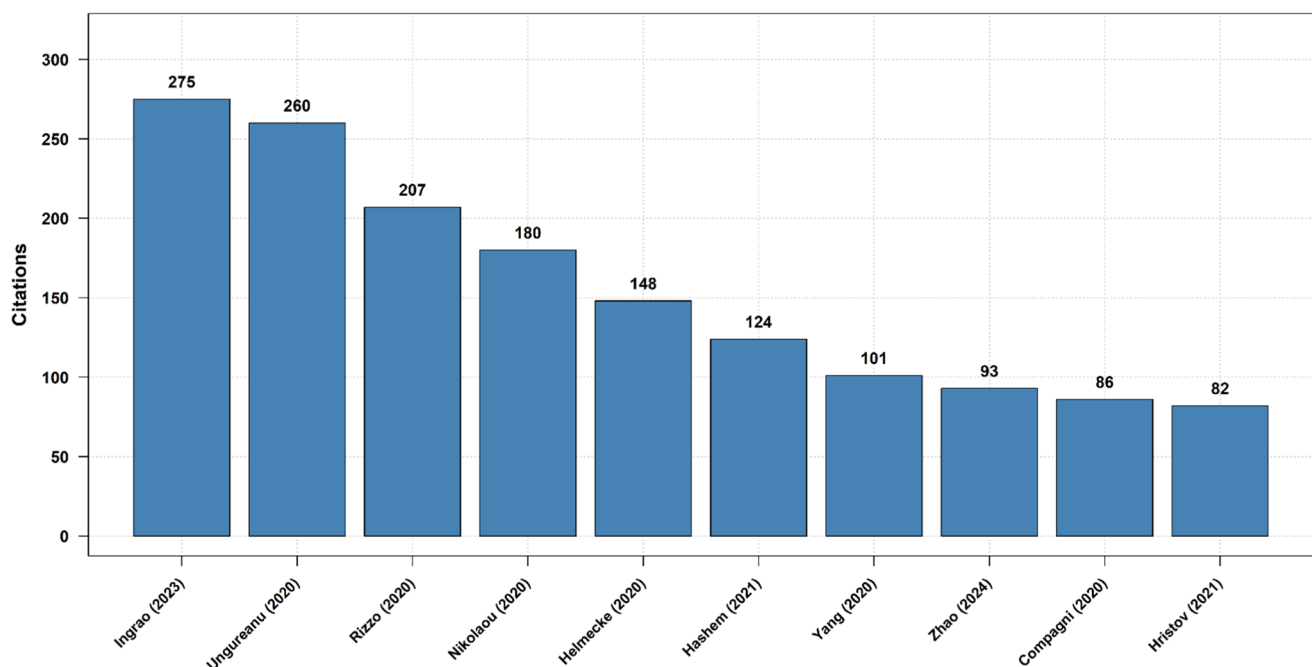


Fig. 6 Most cited publications in research on scientific developments in the use of non-conventional resources

Research Areas

Figure 7 summarizes the eight main Web of Science research areas represented in the 2020–2024 corpus: Agriculture (38%), Environmental Sciences (24%), Science and Technology (11%), Plant Sciences (9%), Engineering (7%), Other categories (7%), Water Resources (3%), and Chemistry (2%). Percentages may not sum to 100% because values were rounded to whole numbers.

Agriculture and Environmental Sciences together accounted for 62% of the mapped areas, confirming a strong focus on crop response, irrigation management, soil–plant interactions, contamination, and water quality. Science and Technology and Plant Sciences added another 20%, emphasizing physiological response and technological development.

Engineering, Water Resources, Chemistry, and the aggregated Other category represented smaller shares. The distribution is interdisciplinary but remains strongly biophysical; public health, economics, governance, and farmer behaviour are less prominent. This imbalance supports studies that follow the same water source through treatment, irrigation, soil, crop, food-safety, groundwater, cost, and adoption outcomes.

Keyword Analysis

Keyword co-occurrence revealed three major conceptual clusters (Fig. 8). Node size reflects term frequency, spatial proximity indicates association, and colours represent clusters; the temporal overlay and total link strength panels show recency and connectivity.

The red cluster centred on soil, irrigation, wastewater, pollution, water quality, health risks, and contaminants, linking reuse to environmental management [4, 37]. The blue cluster captured agronomic performance under salinity through yield, water productivity, evapotranspiration, deficit irrigation, and crop-specific terms [23]. The green cluster emphasized physiological responses to salt stress, including oxidative stress, proline, chlorophyll, photosynthesis, and gas exchange [45, 50].

Across clusters, the strongest bridges remain biophysical. Governance, farmer adoption, treatment cost, lifecycle assessment, public health, groundwater, standardized monitoring, contaminant mixtures, and antibiotic resistance are comparatively less visible. The network therefore demonstrates conceptual coherence around the soil–water–plant

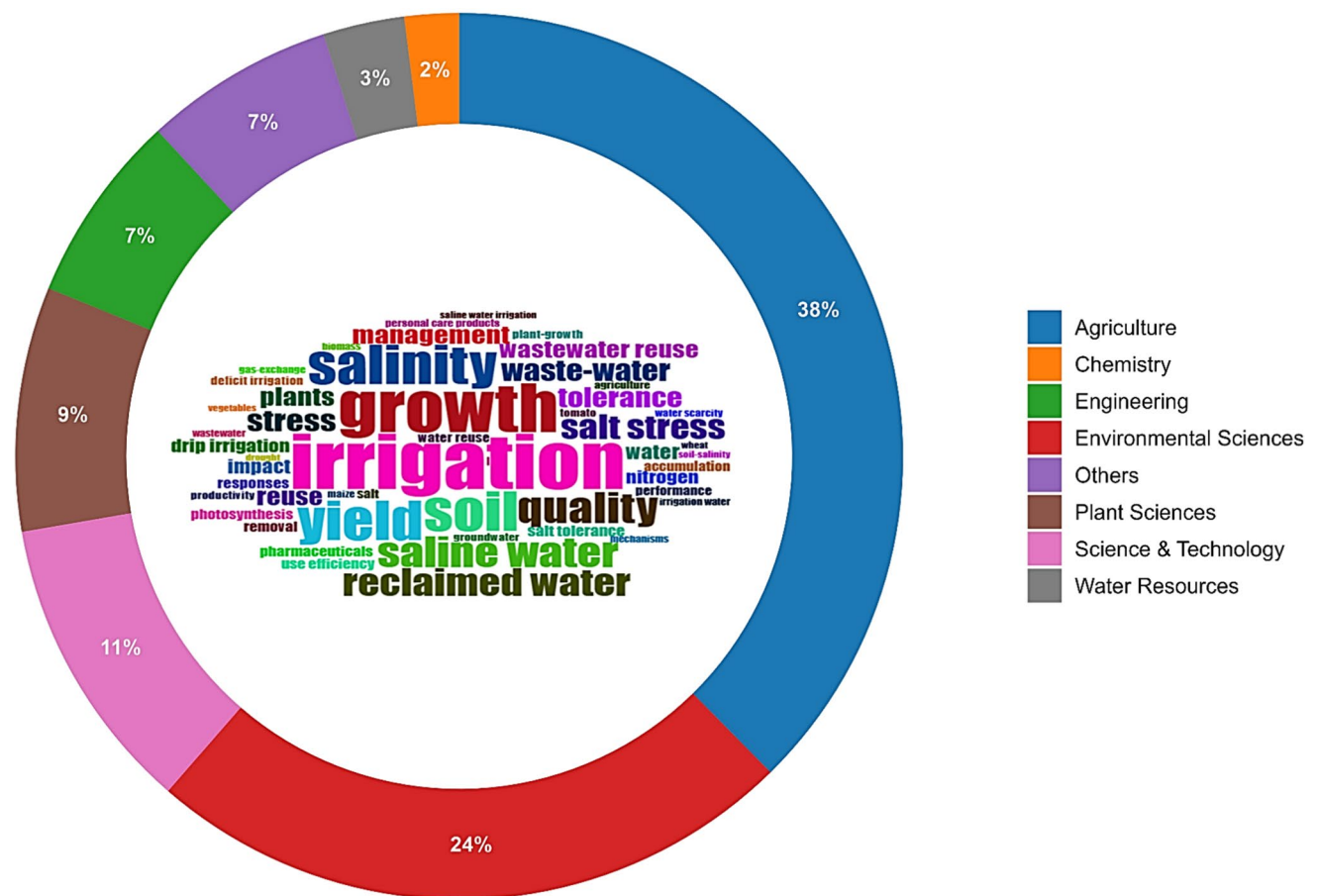


Fig. 7 Distribution of the eight main Web of Science research areas represented in studies on non-conventional water resources for irrigation (2020–2024). Percentages are rounded and may not sum to 100%

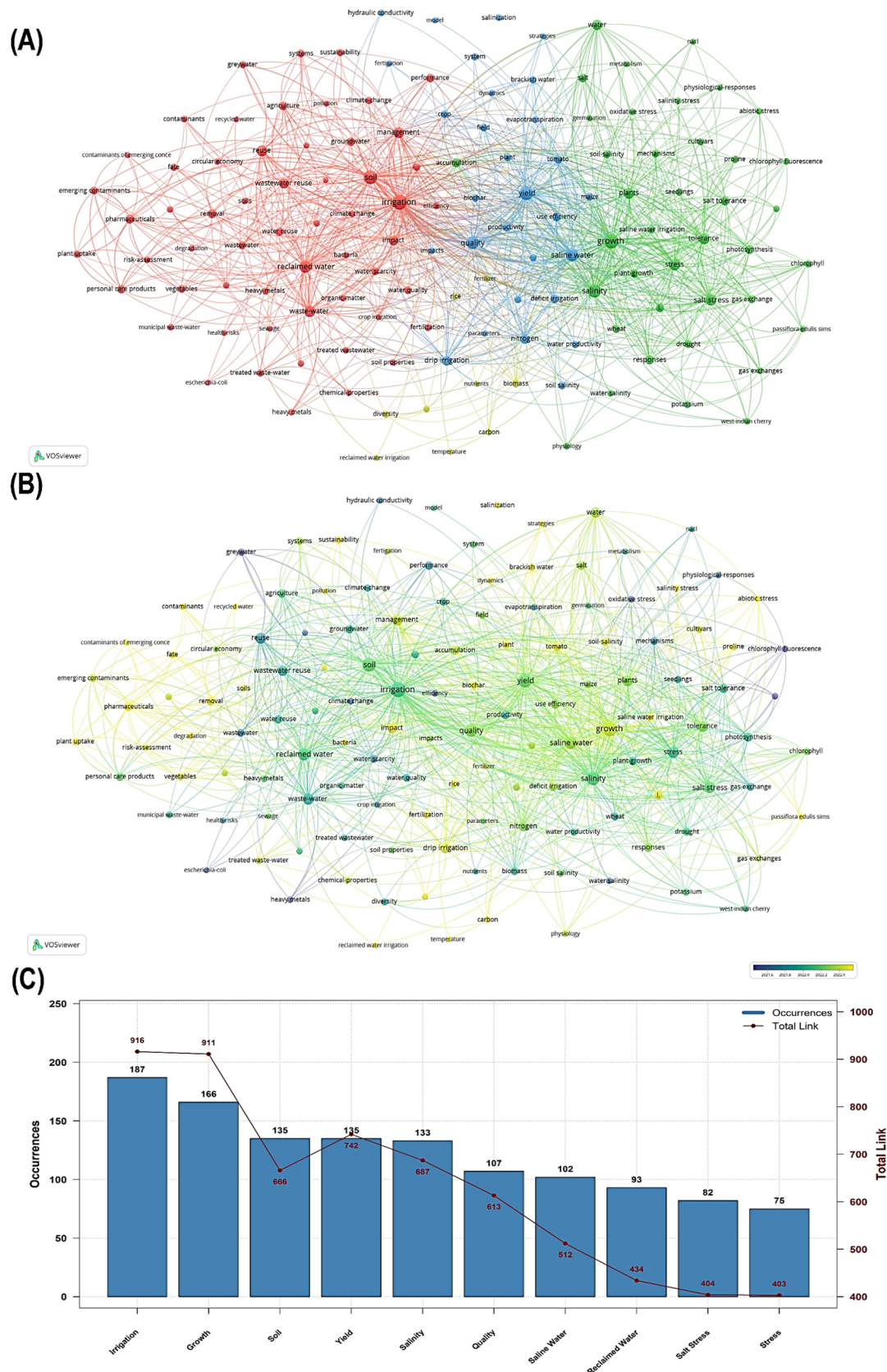


Fig. 8 Keyword co-occurrence structure in research on non-conventional water resources for irrigation. **A** Author-keyword network and thematic clusters, with node size proportional to occurrence frequency.

B Temporal overlay based on average year of occurrence. **C** Keyword connectivity measured by total link strength

system but not equivalent maturity in implementation and risk governance.

The temporal overlay suggests persistence of foundational terms such as salinity, irrigation, soil, and water quality, with later emphasis on efficiency, management, mitigation, sustainability, tolerance, organic matter, and biochar. Because the period spans only five years, this pattern indicates shifting emphasis rather than a definitive thematic transition.

Figure 9 complements the co-occurrence map by showing the conceptual position and maturity of themes. Wastewater reuse, irrigation, and water reuse occupy central positions, whereas salinity/salt stress and crop-oriented physiological themes remain partly separated. This separation exposes a continuing gap between source-quality/contaminant research and plant-stress/agronomic research.

The thematic map is dominated by broad basic themes such as growth, yield, and salinity, while specialized themes remain peripheral. A priority is therefore to connect water-quality profiles and treatment performance with threshold-based soil, crop, groundwater, and food-safety outcomes so that thematic activity becomes decision-relevant evidence.

Critical Synthesis of Evidence, Trade-offs, and Research Gaps

Bibliometric growth and global collaboration coexist with an uneven evidence base. The literature is strong in describing individual hazards, short-term crop responses, and technical options, but weaker in integrating them into source-to-farm-to-food risk frameworks. The critical issue is conditionality: benefits can be substantial when water quality, treatment, soil buffering, crop sensitivity, irrigation design, and monitoring are aligned; the same practice can become degradative when one component is neglected. The following synthesis therefore distinguishes recurrent findings from unresolved uncertainties and implementation gaps.

Non-conventional Water Sources: Definitions and Typologies

The scarcity of fresh water, a vital resource for the health of ecosystems and human survival, has intensified globally, with demand often exceeding availability [35]. Given this scenario, the development and use of non-conventional water resources has become a crucial strategy to mitigate scarcity and reduce water conflicts [31]. These are defined as alternative sources to traditional freshwater and include saline/brackish water, desalinated water, and reclaimed wastewater [9]. The reuse of treated wastewater for purposes such as agricultural irrigation, for example, is recognized as a significant and reliable source of water for irrigation [46].

The types of non-conventional water resources are vast, but can be broadly divided into effluent sources and naturally saline sources. Effluent sources originate predominantly from the urban/domestic and industrial sectors [25]. The domestic sector, in turn, generates urban wastewater and, in particular, gray water (sewage from washing machines, dishwashers, and bathtubs, excluding human waste), which accounts for between 50 and 80% of the total volume of wastewater worldwide [48]. The agricultural sector is the largest consumer of fresh water in the world and, although it is not the main generator in terms of municipal sewage volume, it produces effluents such as agricultural drainage water and agricultural production and processing effluents [29].

The central problem with effluent reuse lies in the presence of various contaminants that persist after conventional treatment [34]. Treated effluents are pathways for the introduction of contaminants of emerging concern into the agroecosystem [12, 28]. Conventional treatment plants are not very effective at removing most contaminants, including pharmaceuticals and personal care products such as the anti-inflammatory diclofenac, the antibiotic sulfamethoxazole, and carbamazepine [20, 43, 44]. The presence of these compounds in soil and crops raises concerns that they may be absorbed and enter the food chain, posing risks to human and environmental health [24].

In addition to chemical contaminants, the reuse of treated wastewater can be a source of antibiotic-resistant bacteria and antibiotic resistance genes, a global health problem exacerbated by the reuse of effluents [17]. Another highly relevant issue is the direct biological risks, especially if the effluent is not properly treated (raw or insufficiently treated domestic wastewater). These effluents are associated with health risks arising from the presence of pathogens [1]. Such pathogens include bacteria (such as *Escherichia coli*), enteric viruses (such as Norovirus and Enterovirus), and protozoa (such as *Cryptosporidium* and *Giardia*) [11]. The contamination of foods, such as leafy vegetables, from crops irrigated with wastewater is a route of transmission for these pathogens [30].

Finally, another type of non-conventional source that is indispensable in arid and semi-arid regions is saline/brackish water. The main problem associated with these waters is salinity, which induces osmotic stress and ionic toxicity in plants, limiting agricultural productivity [33]. Long-term use of saline water for irrigation causes salinity to accumulate in the soil, negatively affecting its physicochemical properties and microbial communities. The main phytotoxic ions are sodium (Na^+) and chloride (Cl^-) [26, 27]. In addition, some industrial effluents, such as wastewater from nickel plating, introduce heavy metals (e.g., nickel (Ni), chromium (Cr), cadmium (Cd), lead (Pb)) that bioaccumulate in the soil and may pose a risk to human health [5, 18].

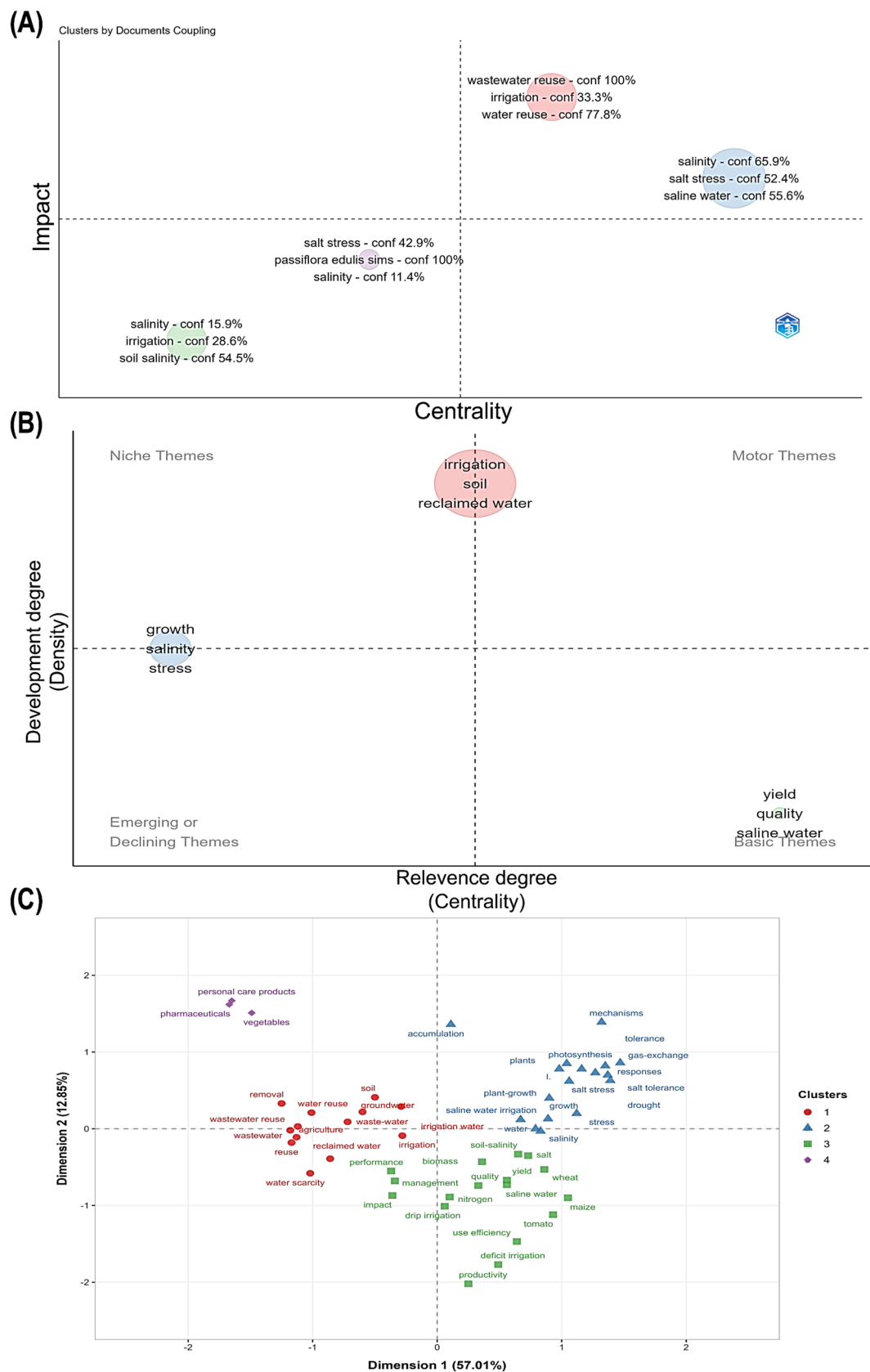


Fig. 9 Conceptual structure of research on non-conventional water resources for irrigation. **A** Document-coupling cluster map positioned by centrality and impact. **B** Thematic map positioned by centrality

and density. **C** Factorial map showing conceptual clusters in reduced-dimensional space

The broad label "non-conventional water" should not be treated as a single exposure category. Reclaimed municipal effluent, produced water, saline groundwater, desalinated water, drainage water, and greywater differ in salinity, nutrients, organic micropollutants, metals, pathogens, and temporal variability. Agriculture-derived contaminants can also reach groundwater through leaching and drainage pathways [36], reinforcing the need to report a minimum water-quality dataset and justify the transferability of conclusions across source types.

Non-conventional Water Quality and its Critical Parameters

Non-conventional water quality is variable rather than intrinsically inadequate. Electrical conductivity and sodium adsorption ratio remain essential for evaluating salinity and sodicity, but they are insufficient as stand-alone indicators. Risk also depends on pH, alkalinity, boron, chloride, nutrient balance, trace elements, suspended solids, pathogens, and source-specific organic compounds, together with crop sensitivity and soil hydraulic properties [2, 21].

The greatest analytical uncertainty concerns contaminants of emerging concern and contaminant mixtures. Heavy metals are persistent and may accumulate, while pharmaceuticals, personal-care products, pesticides, endocrine-active compounds, transformation products, and microplastics can enter coupled water-soil systems. Microplastics can also act as carriers for metals, organic contaminants, nutrients, and antibiotics in freshwater environments [19], although irrigation-specific transfer and long-term crop exposure remain insufficiently quantified. Most irrigation studies still assess selected compounds individually, despite real effluents containing mixtures whose additive, synergistic, or antagonistic effects are poorly resolved.

Long-term assessment must extend beyond irrigation-water compliance to the entire exposure pathway. Baseline and repeated measurements of soil salinity, sodicity, structure, microbial indicators, contaminant pools, crop tissues, drainage, and groundwater are required to distinguish transient responses from cumulative change. Current evidence is often limited by short experimental duration, inconsistent reporting of irrigation loads, and the absence of matched freshwater controls, reducing comparability among studies.

Regulations and treatment technologies should therefore follow a fit-for-purpose approach: the required treatment level must be matched to the water source, crop, edible plant part, irrigation method, soil vulnerability, and exposure route. Advanced processes such as filtration, ultraviolet disinfection, ozonation, membranes, or combined treatment trains can reduce specific hazards, but may increase energy use, costs, concentrate production, or

operational complexity [22, 32]. Safety cannot be inferred from treatment sophistication alone, it must be verified through performance monitoring and clearly defined action thresholds.

Regulatory comparability also remains limited. The European Union applies Regulation (EU) 2020/741, which links reclaimed-water quality classes to crop category and irrigation method; the United States relies largely on state-specific requirements supported by federal reuse guidance; China applies the national GB 5084–2021 Standard for irrigation water quality; and Brazil uses federal resolutions establishing general non-potable and agricultural/forestry reuse directives, complemented by subnational requirements [6, 7, 10, 38, 40]. Middle Eastern frameworks are likewise country-specific. Across jurisdictions, conventional microbiological and physico-chemical criteria are more standardized than thresholds for emerging contaminants, antibiotic-resistance determinants, microplastics, or contaminant mixtures, limiting direct global comparison.

Environmental Impacts: Risks and Opportunities

The environmental case for non-conventional water is inherently conditional. Reuse can reduce freshwater extraction and recover nutrients, but the same nutrient and salt loads can become pollutants when application exceeds crop demand or soil assimilative capacity. A critical assessment must therefore compare avoided impacts with newly introduced risks rather than assuming that resource circularity is automatically sustainable.

Evidence is strongest for salinity, sodicity, and deterioration of soil physical properties under prolonged or poorly managed irrigation. Concerns about trace elements, pharmaceuticals, pathogens, antibiotic-resistance determinants, and groundwater contamination are scientifically credible, but field evidence is uneven across compounds, soils, crops, and exposure durations [3, 17]. The principal gap is not the identification of hazards, but quantification of when cumulative loads cross agronomic, ecological, or health-relevant thresholds.

Potential benefits are also context dependent. Treated wastewater can substitute freshwater and supply nitrogen, phosphorus, and organic matter, while desalinated or saline-water strategies may stabilize production where conventional sources are unavailable. However, nutrient recovery can be offset by nutrient imbalance, salinity management, treatment energy, concentrate disposal, and infrastructure requirements. Claims of carbon sequestration or climate mitigation therefore require lifecycle evidence rather than inference from water reuse alone.

Sustainable implementation requires an explicit decision rule: adoption is justified only when water savings and agronomic benefits are demonstrated together with verifiable control of soil, crop, groundwater, and food-safety risks. Source-specific standards, baseline characterization, cumulative-load accounting, monitoring thresholds, and contingency measures are more defensible than generic claims that non-conventional water constitutes a regenerative solution.

Strategy for Agricultural Use Management and Minimization of Environmental Risks

Management must be designed around the specific water–soil–crop combination. Crop selection, blending, soil amendments, irrigation scheduling, and leaching can reduce salinity or sodicity risks, but each intervention has trade-offs. Drip and subsurface systems may reduce contact between reclaimed water and edible plant surfaces, yet they can also concentrate salts near emitters and remain vulnerable to clogging. No irrigation method is universally safer without considering water quality, maintenance, rainfall, drainage, and root-zone salt balance.

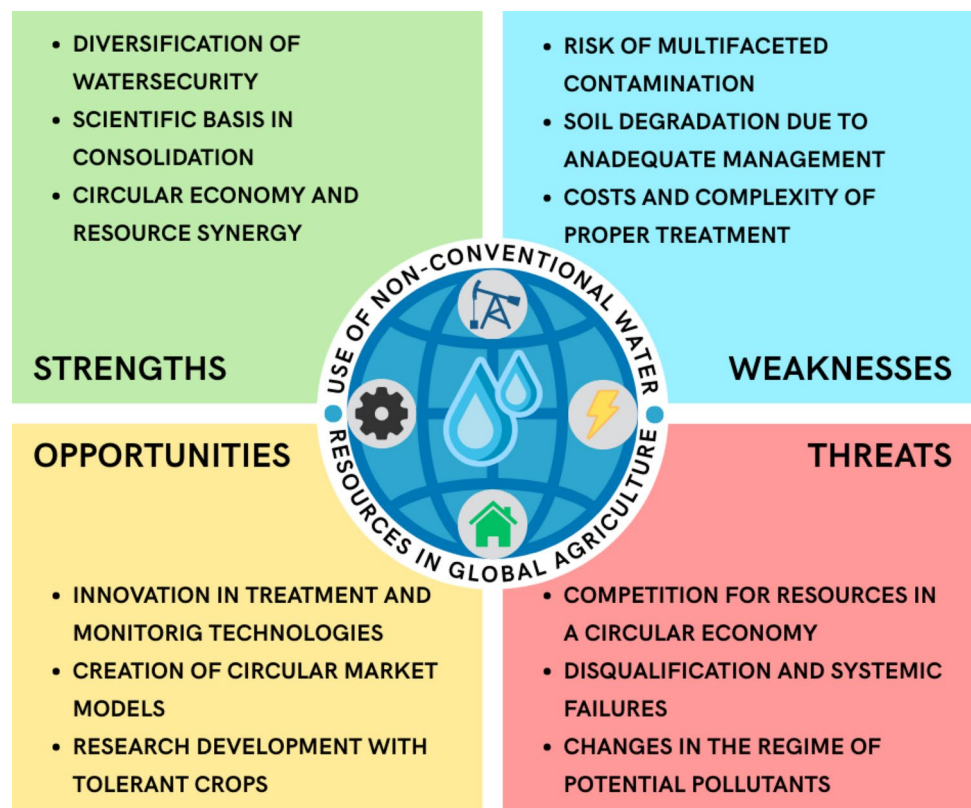
Monitoring should begin before irrigation and continue at intervals defined by cumulative water and solute loads. At minimum, programmes should track irrigation-water

quality, soil electrical conductivity and sodium adsorption indicators, infiltration or structural change, relevant contaminants, crop tissue, and—where leaching is plausible—drainage or groundwater. Corrective actions, including source blending, leaching, amendment application, crop rotation, or temporary suspension, should be linked to pre-determined thresholds rather than implemented only after visible damage.

Governance is as important as treatment. Risk-based standards should define permitted uses, responsibilities for monitoring, data transparency, liability, and responses to non-compliance. Farmer training and extension are essential, but adoption will remain limited if recommendations ignore operating costs, access to laboratories, infrastructure maintenance, market acceptance, and the distribution of risks and benefits along the reuse chain.

Circular water systems can be valuable at local or regional scale, but they are not inherently sustainable. Their performance must account for treatment reliability, energy demand, greenhouse-gas emissions, sludge or concentrate management, transport distance, nutrient substitution, and long-term soil effects. The SWOT analysis in Fig. 10 should therefore be read as a structured hypothesis map that identifies conditions to be tested, rather than as proof that all reuse arrangements deliver net environmental benefits.

Fig. 10 SWOT analysis (strengths, weaknesses, opportunities, and threats) of the use of non-conventional water sources in agricultural irrigation



Prioritized Research Agenda and Implementation Pathway

The first research priority is to replace isolated short-term demonstrations with harmonized, multi-season and multi-site field trials. Experiments should compare treated wastewater, saline/brackish water, desalinated water, harvested rainwater or other locally relevant sources against freshwater controls across contrasting soils, crops, climates, and irrigation systems. Core outcomes should include yield and water productivity together with cumulative salt and nutrient loads, soil structure, crop quality, contaminant uptake, and drainage or groundwater response. Such designs are required before regional differences in long-term agronomic performance or drought resilience can be ranked reliably.

The second priority is integrated One Health risk assessment. Studies should evaluate mixtures of emerging contaminants, microplastics, pathogens, antibiotic-resistant bacteria, and resistance genes across the water-soil-microbiome-crop-groundwater continuum. Research must move beyond presence/absence toward dose–response relationships, transformation products, bioavailability, cumulative exposure, and thresholds relevant to farmers, consumers, ecosystems, and groundwater protection, while testing how soil buffering, climate, crop species or cultivar, and irrigation practice modify mobility and exposure.

The third priority is fit-for-purpose technology and monitoring assessment. Treatment trains, blending strategies, irrigation methods, sensor networks, Internet of Things platforms, remote sensing, and predictive models should be validated under operational variability rather than only under controlled conditions. Evaluation should include removal and exposure reduction, reliability during failures or shock loads, calibration and analytical uncertainty, clogging and root-zone salt balance, maintenance, energy and chemical demand, residual management, lifecycle impacts, and cost per unit of safely delivered irrigation water.

The fourth priority is implementation and comparative sustainability. Cross-jurisdictional studies should compare regulatory thresholds, institutional responsibilities, financing, farmer knowledge and trust, consumer acceptance, liability, and enforcement while pairing these factors with lifecycle and techno-economic analysis of competing water-supply options. The objective is not to maximize reuse, but to identify combinations of water source, treatment, irrigation method, crop selection, monitoring, and governance that remain safe and economically viable under local scarcity conditions. Transition from experimental to operational practice should require multi-season evidence of stable crop performance, controlled soil and groundwater risk, treatment reliability, monitoring capacity, and acceptable lifecycle cost. The evidence gaps and corresponding actions are summarized in Table 1.

Study Limitations and Methodological Considerations

This study used only the Web of Science Core Collection. That choice supports reproducible citation-network analysis but excludes literature indexed only in Scopus, SciELO, regional databases, institutional repositories, and grey literature. Because non-conventional water management is strongly context dependent, underrepresentation of regional evidence may bias the apparent geography, terminology, and priorities of the field.

Retrieval depended on the selected search string and the 2020–2024 period. Although the query captured major reclaimed-water, wastewater, greywater, saline-water, and non-conventional-water terms, it may have missed source-specific expressions such as produced water, drainage water, aquaculture effluent, stormwater, or alternative indexing conventions. The short interval is appropriate for recent dynamics but cannot establish long-term historical shifts.

Bibliometric indicators quantify productivity, citation visibility, and network structure; they do not assess risk of bias, methodological rigor, or effect magnitude. The critical synthesis was not a formal systematic review of every full text and did not apply a standardized quality appraisal or meta-analysis. Consequently, the research priorities identify recurrent evidence gaps but do not provide pooled estimates of treatment efficacy, crop response, or health/environmental risk.

Citation lag, author and institution name variants, database normalization, threshold choices, and the absence of field- and year-normalized citation metrics can influence rankings and network geometry. Spatial proximity, node size, and link strength do not demonstrate causality, quality, or practical effectiveness. In addition, the search strategy did not explicitly cover every possible non-conventional source term, so conclusions should not be generalized beyond the evidence represented by the retrieved corpus.

Conclusions

Global research on non-conventional water for irrigation expanded during 2020–2024, but the field is better described as scientifically active than uniformly mature. Publication and collaboration networks demonstrate visibility and international participation; they do not by themselves establish evidence quality, transferability, or safety.

The literature supports opportunities for freshwater substitution, nutrient recovery, and drought adaptation, while also documenting risks from salinity, sodicity, emerging contaminants, pathogens, antibiotic resistance, soil degradation, and groundwater exposure. These outcomes are

Table 1 Critical evidence gaps and priority recommendations for research and implementation of non-conventional water in irrigation

Evidence domain	Critical limitation	Why it matters	Priority research action
Water-source characterization	Inconsistent reporting and aggregation of chemically distinct water sources	Limits comparability and can transfer conclusions across incompatible hazard profiles	Adopt a minimum dataset for salinity, sodicity, nutrients, trace elements, pathogens, and source-specific organic contaminants; report temporal variability
Long-term soil and crop response	Predominance of short-term, single-site, pot, greenhouse, or single-season experiments	May miss cumulative salinization, structural degradation, contaminant accumulation, and delayed yield or quality effects	Conduct multi-season, multi-site field trials with freshwater controls and cumulative load accounting across contrasting soils, crops, and climates
One Health and mixture risk	Fragmented assessment of individual contaminants, pathogens, and resistance determinants	Real waters contain mixtures and expose soils, microbiomes, crops, farmers, consumers, and groundwater simultaneously	Evaluate mixtures, transformation products, bioavailability, dose–response, ARGs/ARB, pathogens, and food-chain exposure in integrated designs
Treatment and irrigation design	Removal efficiency is often evaluated without reliability, residuals, clogging, root-zone salt balance, or failure scenarios	A technically advanced process may shift impacts to energy use, brine, sludge, cost, or maintenance	Use fit-for-purpose treatment and irrigation comparisons that include operational reliability, residual management, energy, maintenance, and action thresholds
Monitoring and digital tools	Sensors, IoT, remote sensing, and predictive models are frequently demonstrated under controlled conditions	Unvalidated data streams can create false confidence and delayed corrective action	Validate accuracy, calibration, uncertainty, interoperability, failure detection, and decision rules under commercial farm conditions
Regulation, economics, governance, and adoption	Fragmented regulatory thresholds and limited evidence on costs, responsibilities, farmer trust, consumer acceptance, liability, and enforcement	Technical feasibility does not ensure cross-jurisdictional comparability, adoption, equity, or sustained compliance	Compare regulatory thresholds across jurisdictions; integrate techno-economic analysis, implementation research, extension, risk communication, financing, market response, and enforceable governance
Comparative sustainability	Few matched long-term comparisons among reclaimed wastewater, desalinated water, saline/brackish water, rainwater harvesting, and freshwater controls	Water savings may be offset by treatment energy, transport, concentrate disposal, salinity management, or long-term soil damage	Apply lifecycle and multi-criteria assessment across water sources, climates, crops, and irrigation systems, including drought resilience and operational costs

conditional on water source, treatment, soil, crop, irrigation method, cumulative loading, monitoring, and governance rather than universally beneficial or harmful.

The principal gap is the limited integration of long-term field performance with soil change, crop uptake, food safety, groundwater protection, lifecycle impacts, costs, regulation, and adoption. Priorities therefore include harmonized reporting, multi-season comparative trials, mixture-based One Health assessment, fit-for-purpose treatment, operational validation of monitoring technologies, lifecycle and techno-economic comparison, and risk-based regulation. Non-conventional water should transition from experimental use to routine agricultural practice only where local evidence demonstrates sustained

agronomic benefit together with controllable environmental and health risks and feasible monitoring and corrective capacity.

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Declarations

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